

Report
Of
JAVA FULL STACK DEVELOPMENT

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number: CS-2341597

Name of Student: ANUJ SINGH

Batch: 2023-27

CANDIDATE'S DECLARATION

I, Anuj Singh do hereby declare that the internship report titled **“Java Full Stack Developer Virtual Internship”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature :-

A handwritten signature in blue ink that reads "Anuj". The letter 'A' is large and stylized, with a horizontal line underneath it. The rest of the name "nuj" is written in a cursive style.

Student Name: Anuj Singh
Roll No: CS-2341597

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature :-

A handwritten signature in blue ink that reads "Anuj". The letter 'A' is large and stylized, with a horizontal line extending from its base. The 'n' and 'j' are written in a cursive, connected style.

Student Name: Anuj Singh
Roll No: CS-2341597



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



Certificate of Virtual Internship

This is to certify that

anuj singh

IILM University, Greater Noida

has successfully completed the 8-weeks

Java Full Stack Development With Project Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported by



EduSkills
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Certificate ID : 4554be59aaa35039b629

Student ID: STU69971ec2b4f831771511490



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

The internship was an eight-week learning program in Java Full Stack Development, undertaken as part of the AICTE – EduSkills Virtual Internship Program during June – August 2026. As recorded on the completion certificate, the program covers the complete development path, starting from the design of a web page's user interface and extending through server-side application logic to storage and retrieval of data in a relational database.

Rather than treating each technology in isolation, I approached the internship as a single, connected learning journey. My aim was to understand how a front-end interface communicates with back-end logic, how that logic in turn interacts with a database, and how the resulting codebase can be maintained through disciplined version control.

The internship helped me link ideas that are usually taught as separate college subjects. Page structure and presentation (HTML/CSS) sit alongside client-side interactivity (JavaScript), Java forms the core programming layer, Spring and Spring Boot support enterprise-grade back-end development, Hibernate bridges Java objects with relational tables, and MySQL provides persistent storage – with Git tying the whole workflow together.

4.2 Organization / Internship Profile

The internship was delivered under the AICTE – National Internship Portal initiative, supported by EduSkills Academy, and endorsed by the Ministry of Education. The certificate of completion confirms the internship domain, duration, and the grade obtained; it does not list an individual mentor or a physical workplace, since the program was conducted in virtual mode.

Field	Detail
Internship Domain	Java Full Stack Developer Virtual Internship
Duration	8 Weeks (June – August 2026)
Mode	Virtual Internship (AICTE – EduSkills, National Internship Portal)
Certificate ID	4d8195f68cc996e0b714
Student ID	STU66446e214bd6b1715760673
Grade Awarded	E (Excellent: 80–89)
Supporting Organization	EduSkills Academy
Institute Name	IILM University, Greater Noida

4.3 Problem Statement

A modern web application has to bring together an approachable interface, dependable business logic, and structured data storage. It is common for a student to understand HTML, Java, Spring, and MySQL individually while still struggling to see how they operate as one working system. This internship was designed to close that gap by building a practical, layer-by-layer understanding of a Java-based full-stack application.

Stated directly, the problem this internship addressed was: how does a developer design a web application in which the front end delivers a smooth user experience, the Java back end processes requests and business rules, and the database stores and serves data reliably – while the whole codebase remains maintainable over time? Git and version control were included specifically to address that last part of the problem.

4.4 Project Objectives

1. To understand the structural building blocks of a web page using HTML.
2. To apply CSS fundamentals and build responsive layouts with Bootstrap.
3. To implement client-side interactivity using JavaScript and jQuery.
4. To build programming fundamentals through Core Java.
5. To explore Advanced Java concepts relevant to enterprise applications.
6. To understand how the Spring Framework structures Java enterprise development.
7. To learn how Spring Boot simplifies configuration and application development.
8. To learn Hibernate and the fundamentals of Object-Relational Mapping (ORM).
9. To use MySQL for structured, relational data storage.
10. To practice source-code management using Git and version control.
11. To integrate all of the above into one coherent full-stack development workflow.

4.5 Scope of the Project

The scope of this report is limited to the Java Full Stack Development learning areas confirmed by the internship certificate. It covers the concepts and practical understanding gained across front-end development, core and advanced Java programming, back-end development with Spring, database interaction, and version control.

Area	Scope
Front End	HTML, CSS, Bootstrap, JavaScript and jQuery for structure, styling, responsiveness and interaction.
Programming	Core Java and Advanced Java as the foundation for application development.
Back End	Spring and Spring Boot for Java-based enterprise application development.

Data Layer	Hibernate for ORM concepts and MySQL for relational data storage.
Development Process	Git and version control for tracking and managing source-code changes.

As the certificate supplied for this report does not name a specific client, employer, or production deployment, this report does not claim any commercial application – it focuses on the learning and development scope confirmed by the internship credential itself.

4.6 Technologies and Tools Used

Technology / Tool	Purpose / Learning Area
HTML	Structure and semantic organization of web pages.
CSS	Styling, layout, spacing, typography and presentation.
Bootstrap	Responsive design framework for adapting interfaces to different screen sizes.
JavaScript	Client-side logic and interactive page behaviour.
jQuery	Simplifying common JavaScript-based DOM and interaction tasks.
Core Java	Programming fundamentals, object-oriented logic and problem solving.
Advanced Java	Enterprise-oriented Java application development concepts.
Spring Framework	Structured, enterprise-grade Java application development.
Spring Boot	Simplified configuration and rapid application development in Java.
Hibernate	Object-Relational Mapping between Java objects and relational data.
MySQL	Relational database for storing and managing structured application data.
Git	

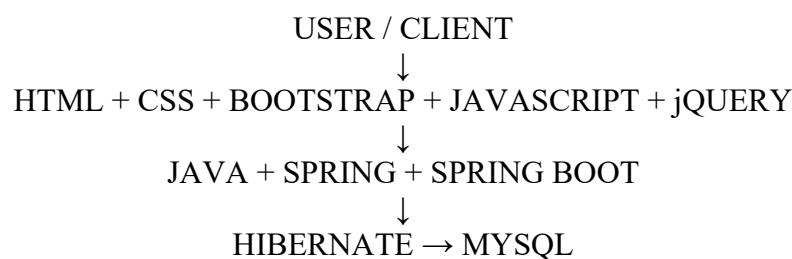
4.7 System Architecture

Source-code version control and tracking of development changes.

The technologies covered in the internship map onto a layered full-stack architecture: a presentation layer that the user sees and interacts with, an application layer that processes requests and business logic, a persistence layer that connects the application to stored data, and a database layer that holds that data. Git supports the development process across all of these layers.

Layer	Main Technologies	Role
Presentation Layer	HTML, CSS, Bootstrap, JavaScript, jQuery	Page structure, design, responsiveness and interaction.
Application Layer	Java, Spring, Spring Boot	Programming logic, application flow and server-side development.
Persistence Layer	Hibernate	Maps application objects to relational data.
Database Layer	MySQL	Stores and manages structured application data.

Figure 4.1: Conceptual architecture of the Java Full Stack learning environment.



This is a conceptual diagram prepared from the technologies listed on the internship certificate; it does not represent any specific proprietary or production system.

4.8 Methodology

The learning process moved progressively from web fundamentals toward Java-based enterprise concepts, so that each new layer built on the one before it.

Phase 1 – Web Fundamentals

Learning began with HTML page structure, followed by CSS fundamentals and Bootstrap for responsive design.

Phase 2 – Client-Side Interaction

JavaScript introduced interactive page behaviour, and jQuery was used to simplify common DOM-handling tasks.

Phase3 –JavaProgramming

Core Java built the programming foundation needed for the enterprise topics that followed.

Phase4 –Enterprise Java

Advanced Java concepts introduced enterprise application development and the motivation for using frameworks.

Phase 5 – Spring and Spring Boot

The Spring Framework and Spring Boot were studied as tools that structure and simplify Java enterprise development.

Phase6 –DatabaseIntegration

Hibernate introduced ORM concepts, and MySQL served as the relational database connecting application logic to persistent data.

Phase 7 – Version Control

Git and version control were used to track, organize and maintain source code as the application evolved.

LearningWorkflow

1. Understand the concept.
2. Study the syntax or framework structure.
3. Apply the concept through small development exercises.
4. Identify errors and debug them.
5. Connect the concept with the next layer of the application.
6. Review the implementation and improve understanding.
7. Track development changes using version control where applicable.

Practical Understanding Developed

One of the most valuable outcomes of the internship was seeing how the different technologies relate to each other. Before working through the full stack, it is easy to view HTML, Java, databases and frameworks as unrelated subjects; the internship helped me see them instead as connected stages of one development process – the front end collects user input, application code processes it, Hibernate maps that data to the persistence layer, MySQL stores it, and Git keeps the evolving source code organized.

I also came to appreciate that frameworks build on, rather than replace, programming fundamentals – Spring and Spring Boot are still fundamentally Java, and Hibernate is easier to understand once relational database concepts are already clear.

Challenges and How I Approached Them

Challenge	Approach
Understanding the full stack	Learned the stack layer by layer instead of trying to absorb every technology at once.
Connecting Java with databases	Focused on tracing a single piece of data from the application layer through Hibernate to MySQL.
Learning frameworks	Treated Spring and Spring Boot as an extension of Java rather than as unrelated new tools.

Debugging	Read the error message, identified the affected layer, checked the relevant code, and tested one change at a time.
Version control discipline	Used Git to think of changes as small, trackable units, which clarified why change history matters.

4.9 Expected Outcomes

The internship certificate records successful completion with a Grade of E (Excellent). Based on the technologies covered, the major learning outcomes are summarised below.

1. Ability to explain the basic role of front-end technologies in a web application.
2. Improved understanding of responsive web design using CSS and Bootstrap.
3. Greater confidence in using JavaScript and jQuery for client-side interaction.
4. A stronger foundation in Core Java programming.
5. A basic understanding of enterprise-oriented Java application development.
6. An understanding of the role of Spring and Spring Boot in Java application development.
7. An understanding of ORM concepts through Hibernate.
8. A basic understanding of relational data storage using MySQL.
9. Awareness of the importance of Git and version control in software development.
10. A clearer picture of how front-end, back-end and database technologies work together in a full-stack application.

Personal Learning Reflection

This internship gave me a more realistic sense of what full-stack development actually involves. Rather than thinking of programming as simply writing code, I began to see it as a complete process – designing an interface, handling data, writing application logic, working with a database, testing changes, and keeping the source code organized.

The biggest lesson for me was the value of pacing myself. With so many technologies covered across eight weeks, trying to master everything at once would have been overwhelming; breaking the material into weekly, layered modules made it manageable, and I became more comfortable treating debugging as a normal, expected part of development rather than a setback.

Going forward, I intend to keep building on Java, Spring Boot, database design, and general software engineering practice so that I can apply this foundation to larger, real-world projects.

4.10 Certificate of Completion and Other Communication Proof

Java Full Stack Developer Virtual Internship, eight-week duration (June – August 2026), Grade E, Certificate ID 4d8195f68cc996e0b714, Student ID STU66446e214bd6b1715760673, issued under the AICTE – EduSkills Virtual Internship program and countersigned by the Chief Coordinating Officer, AICTE, Ministry of Education, and the Chief Executive Officer, EduSkills. The certificate is reproduced on page 4 of this report.

5. BIBLIOGRAPHY / REFERENCES

1. Certificate of Virtual Internship issued to Trisha Yadav, Java Full Stack Developer Virtual Internship, June – August 2026, Certificate ID: 4d8195f68cc996e0b714.
2. IILM University, School of Computer Science and Engineering – Internship Report Format and Instructions supplied as the reference format for this report.
3. Oracle Java Documentation – reference material for Java programming concepts.
4. Spring Framework Documentation – reference material for Spring concepts and application development.
5. Spring Boot Documentation – reference material for Spring Boot application development.
6. Hibernate ORM Documentation – reference material for Object-Relational Mapping concepts.
7. MySQL Documentation – reference material for relational database concepts.
8. Git Documentation – reference material for version control and Git usage.
9. MDN Web Docs – reference material for HTML, CSS and JavaScript web-development concepts.
10. Bootstrap Documentation – reference material for responsive web design concepts.